

# Sam Ayoub

AI/ML Architecture Director & Engineering Team Lead // Big Data & ML Training Systems // Software Migration & Integration Architect

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## SUMMARY

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Engineering leader and principal architect directing cross-functional teams and technical strategy for AI/ML integration, big data platforms, and machine learning model training. As founder and principal architect of Reallexi LLC, I direct the architecture of a production AI product suite — private RAG memory systems, a local ML training platform with checkpoint-aware fine-tuning and artifact lineage, autonomous engineering agents, generative media pipelines, and a governed API gateway — owning each system end-to-end, from data pipeline to trained model to production inference. Alongside 15+ years architecting FinTech, SaaS, and enterprise platforms, including large-scale API and cloud modernization at First Citizens Bank and Silicon Valley Bank, I bring the same discipline to AI systems: scalable big-data pipelines, reproducible ML training, RAG and vector retrieval, and multi-agent orchestration, all delivered with the security, governance, and team mentorship that production AI demands.

## CORE FOCUS

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- Directing engineering teams and owning end-to-end architecture for AI/ML integration, from big data pipelines to production inference.
- Leading ML model training strategy — dataset pipelines, fine-tuning, checkpoint lineage, and reproducible training operations at scale.
- Architecting big data platforms and pipelines that feed reliable, governed context into LLMs, RAG systems, and autonomous agents.
- Directing cross-functional engineering teams across FinTech, AI, and SaaS products, mentoring engineers and setting technical roadmap.
- Building and shipping production AI systems — RAG platforms, autonomous agents, generative media pipelines, and API infrastructure.
- Partnering with executive leadership to align AI/ML and big-data architecture with product strategy, compliance, and NVIDIA GPU-accelerated computing.

## EXPERIENCE

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### First Citizens Bank, North Carolina

Apr 2023 – Present

#### *Software Architect Consultant*

Directing enterprise-wide architecture strategy and cross-functional engineering teams for scalable API, cloud, and big-data modernization across the FinTech platform. Own the technical roadmap for AI/ML integration — LLM-powered document processing, ML model training and fine-tuning pipelines, AI agent orchestration, and NVIDIA GPU-accelerated inference. Advise on API governance, developer experience, and DevOps automation using GitLab CI/CD while partnering with leadership on hybrid AWS/Azure cloud migration and AI-first architecture strategy.

### Silicon Valley Bank, San Diego

Jul 2020 – Apr 2023

#### *Senior Principal Software Architect*

Directed architecture strategy and technical teams delivering scalable API frameworks with AI-powered analytics, reducing operational downtime by 30%. Pioneered LLM integration for automated compliance monitoring and intelligent API documentation generation. Led multi-functional engineering teams through enterprise platform modernization, introduced API specification standards, and directed big-data pipelines and ML model training/evaluation for fraud detection and risk assessment.

### Achieve Internet, San Diego

Mar 2019 – Mar 2020

#### *Senior Lead Engineer*

Oversaw architecture and DevOps automation for API-driven enterprise applications across AWS, GCP, and Azure. Introduced AI-assisted code review and automated testing pipelines using machine learning models. Directed cloud security and multi-tenant compliance programs while streamlining CI/CD pipelines to improve release efficiency by 35% and implementing intelligent monitoring with anomaly detection.

Automated the development lifecycle with Jenkins and CircleCI, cutting manual deployment effort by 60%. Delivered scalable API integrations that supported accessibility innovations for visually impaired users, including AI-powered visual recognition and natural language processing features. Collaborated with Marketing on SEO initiatives that increased web traffic by 20% and explored early-stage conversational AI prototypes.

Led global development of the Outsell OSdata platform serving over 10 million companies with real-time analytics and early machine learning-driven insights. Designed and deployed SSO authentication systems and coordinated remote engineering teams across time zones to consistently achieve on-time delivery. Introduced data pipeline automation and predictive analytics capabilities for market intelligence.

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## AI/ML PRODUCT ARCHITECTURE — FOUNDER & PRINCIPAL ARCHITECT, REALLEXI LLC

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### RealLexi / RealMemoryAI

[realmemoryai.com](https://realmemoryai.com)

*Private, source-grounded company memory for AI agents*

Directed the architecture of a tenant-isolated company-memory platform that turns documents, URLs, images, and email sources into governed AI context. Owns the full path from ingestion to grounded retrieval to scoped agent and API access, so answers stay traceable to approved sources instead of becoming an invisible permission layer.

### AI Model Builder

[llm.reallexi.io](https://llm.reallexi.io)

*Local ML training with checkpoint lineage and proof*

Architected a local ML training platform that turns ambitious training targets into bounded, recoverable phases. Directed the design of the phased-training engine, checkpoint-aware recovery, and artifact lineage (LoRA adapters, standalone SafeTensors, GGUF) so every training run stays auditable and resumable, with a direct publishing path to the Hugging Face Hub.

### AI Apply Pilot

[aiapplypilot.com](https://aiapplypilot.com)

*Evidence-based job search, human-approved every step*

Architected an AI job-search workspace that compares a candidate's resume against a job description, surfaces real alignment and gaps, and drafts evidence-linked resume updates for human review. Designed the product so AI assists analysis while every submission, approval, and edit stays under the candidate's control.

### Real Memory AI Agent

[marketplace.visualstudio.com/items?itemName=ReallexiLLC.real-memory-ai-agent](https://marketplace.visualstudio.com/items?itemName=ReallexiLLC.real-memory-ai-agent)

*A workspace-aware engineering agent, not a chat panel*

Designed a VS Code engineering agent with Planner + Coder, Auto, Direct, and Single execution modes that inspects and edits the real workspace, runs commands and validation, and preserves session and project evidence — extended across Chrome, CLI, and MCP-assisted workflows.

### RealLexi AI Studio

[ai.reallexi.com](https://ai.reallexi.com)

*Privacy-aware AI image & video generation studio*

Architected a creative studio for generating and editing AI images and video from reference media, with a review step before anything reaches the public gallery — built privacy-first, with generation pipelines routed across multiple model providers.

### RLX Gate

[rlxgate.com](https://rlxgate.com)

*A governed control plane for Envoy — no Kubernetes required*

Architected a self-hostable API gateway that separates a Go-based control plane from the Envoy data plane, so traffic decisions are governed, signed, and auditable before they ever touch production. Supports OpenAPI import with malware scanning, weighted backends with health checks, and API key/JWT/OIDC identity, deployable on a single VPS without a Kubernetes dependency.

### ApiDocPro

[apidocpro.com](https://apidocpro.com)

*API documentation as a hosted, always-current service*

Built a SaaS documentation engine that imports OpenAPI, Swagger, or GraphQL definitions and turns them into modern, branded, hosted docs — including the custom OpenAPI/AsyncAPI UI renderer at [ui.apidocpro.com](https://ui.apidocpro.com) — so engineering, writing, and product teams stay aligned with the real implementation instead of hand-maintained docs.

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## SKILLS & TECHNOLOGIES

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Technical · Computer Science · Innovation · Software Engineering · Software Development · Architecture Design · Java · JavaScript · Git · SQL · NoSQL · Splunk · AngularJS · HTML · Technical solutions · Mentoring · DevOps · Clouds · Apigee · Laravel · PHP · XML · Leadership · Management · SEO · UX/UI · Modernization · CSS · NodeJS · ReactJS/VueJS

## CERTIFICATIONS

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Meta Front-End Developer — Meta

Veracode Security Labs Champion Level 2 — Amazon Web Services (AWS)

API Design and Fundamentals of Google Cloud's Apigee API Platform (with Honors) — Google Cloud Education

Advanced react — Meta

AWS Fundamentals: Building Serverless Applications — Amazon Web Services (AWS)

AWS Fundamentals: Going Cloud-Native — Amazon Web Services (AWS)

AWS Fundamentals: Migrating to the Cloud — Amazon Web Services (AWS)

Continuous Delivery & DevOps — University of Virginia

Veracode Security Labs Champion Level 2 — Veracode

AWS Certified Solutions Architect Associate — Amazon Web Services (AWS)

## EDUCATION

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MBA, Management

Keller Graduate School of Management — 2014

Bachelor of Science, Computer Science

New York Institute of Technology — 2012